Busy counties are big counties. Per resident, food-shelf activity barely follows estimated need.

Across 85 counties, visits, pounds and individuals served track Feeding America's count of food-insecure people almost in lockstep (ρ 0.90–0.92), because both mostly measure population (ρ 0.99).

Per resident the link is weak at best: ρ −0.02 to 0.23 across 2022–24, and none stands out once we allow for testing nine times.

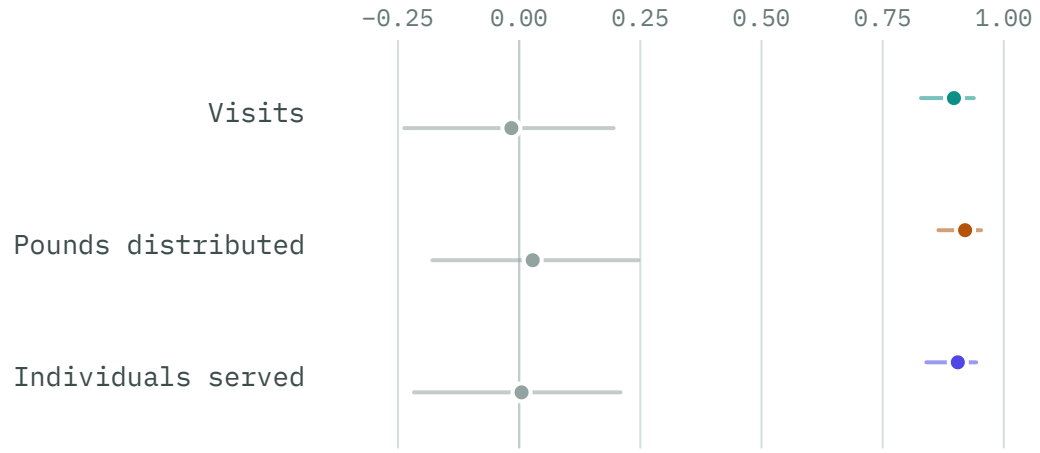
ρ (rho): 1 means two lists rank counties identically, 0 means unrelated.

How closely does activity follow the hunger estimate?

Rank correlation by county, 95% range. Right = in step; 0 = unrelated.

2022 | 2023 | 2024

Total activity vs people (signal colour) | Per resident vs rate



MONDAY Allocation by visits rewards size. Keep a need measure that does not depend on visit volume.

Feeding America, Map the Meal Gap 2024, 2025 (v2), 2026 editions (Dewey et al. 2024; Ribar et al. 2025, 2026) · The Food Group, HouseholdsReg, PoundsReg, IndividualsReg, 2022–24 · Census ACS 2019–23 population

A county's number is often one shelf, and where that shelf is booked can decide the county's rank.

ONE SHELF IS MOST OF THE COUNTY

32 of 86

counties where one shelf carries 75% or more of individuals served (2024)

BOOKED TO ANOTHER COUNTY

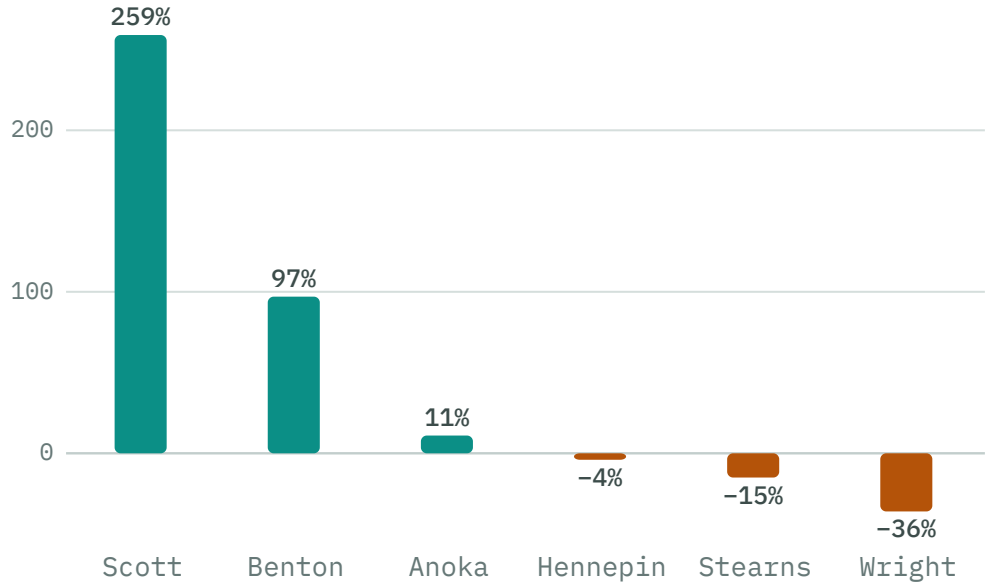
3.2%

of people served are at 21 shelves booked to a different county than their census block

Counting each shelf in its census-block county moves no correlation and keeps 13 of 14 flagged counties. Scott's low activity and Benton's flag each hinge on one shelf.

2024 individuals served, counted by census-block county

Percent change vs the booked county. Hover for counts.



MONDAY Treat a surge or a drop as a reporting question first: confirm each large shelf's county in the master before reading a ranking.

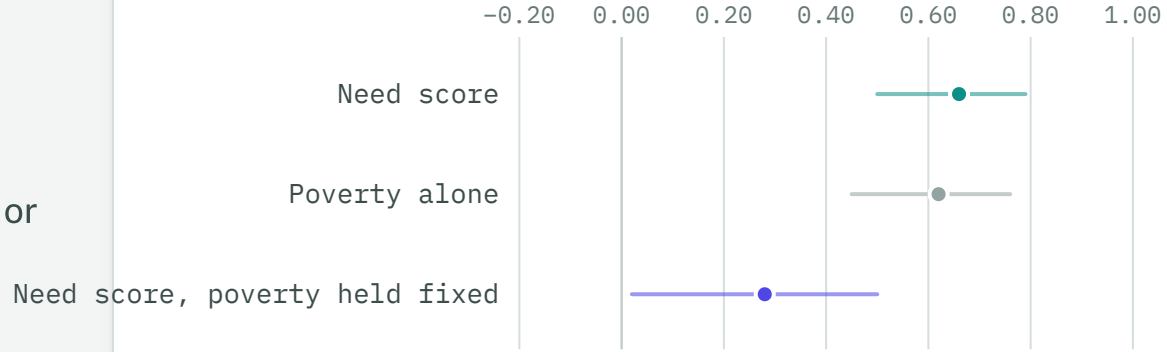
The Food Group org_FSStats and Organization master (census block), IndividualsReg 2024 · Census 2020 block interior points

We define need without any activity data, and we say what would make us reject it.

- **Need score:** average percentile of food-insecurity rate, poverty rate and car-free households. No activity data.
- **Flag:** need above the median, and activity per resident 25+ points lower (0–100 scale) for visits, pounds and individuals.
- **Reject if** there is no rise with SNAP participation, or a survey of non-users (not yet run) finds no more hardship where it is high.
- **Revise if** ranks move with weights or inputs. They do, so we narrow the list.

Does the score rise with SNAP participation?

Rank correlation, 82 counties, 95% range. SNAP is not an input.



The rule **passed**, but poverty alone scores as well. At equal poverty the score still rises, **only just** (0.28, range 0.02 to 0.50).

MONDAY Use the score to decide where to look, not how much food to move: audit first, reallocate second.

Feeding America MMG 2026 (food-insecurity rate) · Census ACS 2019–23 B17001, B08201 · Minnesota DCYF SNAP persons per resident, 2022–24 · poverty alone reaches 0.62, the full score 0.66

6 counties stay on the call list under every test we ran, a list chance alone could produce.

AUDIT FIRST	NEED PERCENTILE	AREA	TESTS PASSED
Chippewa	99	Outside metro	**5 of 5**
Lyon	86	Outside metro	**5 of 5**
Wadena	89	Outside metro	**5 of 5**
Todd	78	Outside metro	**5 of 5**
Swift	70	Outside metro	**5 of 5**
Nobles	93	Outside metro	**5 of 5**

Eight more are flagged but fail at least one test (Clearwater, Aitkin, Benton, Winona, Carlton, Roseau, Norman, Beltrami): *watch*, appendix A3.

A queue, not a verdict. Pair need with activity at random and the same rule flags about 16 counties (11–21), and the weight and input tests still leave about 12 (7–16). We flag 14 and leave 8. The names are a reproducible call list, not proof these counties are unusual.

Tribal programs and tribal-agency SNAP sit outside county rows, so reservation counties read as undercounts.

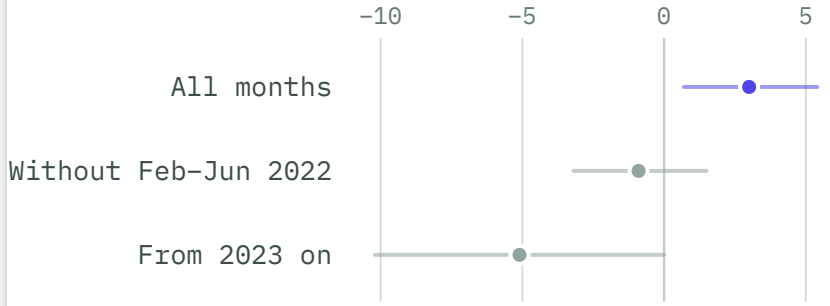
MONDAY Start with a conversation, not a review: hours, transport, language and unreported shelves. Offer help with reporting. Nothing about funding changes until we have listened.

Tests: five activity universes · three other weightings · dropping each input · shelves counted in census-block county · 2,000 redraws within Census margins of error · Feeding America MMG 2026, Census ACS 2019–23, USDA ERS rural-urban codes 2023

Prices give no reliable early warning, and statewide totals cannot isolate what SNAP changes did.

Q4 · individuals served: % change per 1% price rise

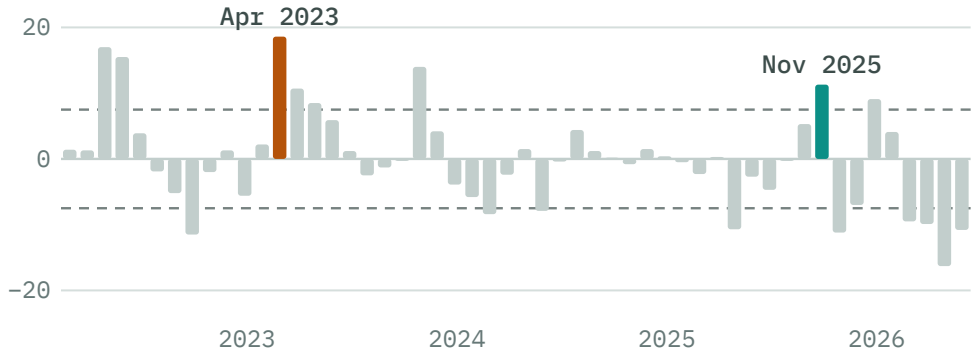
Midwest food-at-home prices, 95% range. Visits and pounds show no clear link; this one rests on early 2022.



SNAP: where a county agency's caseload grew faster, visits grew a little faster (0.27 per 1%, range 0.05 to 0.48): co-movement, not proof that shelves replace benefits.

Q5 · how far does statewide visit activity swing in a month?

Points above or below the usual change for that calendar month. Dashed lines: a typical swing (±7.5).



April 2023 (emergency allotments end) and **November 2025** (benefits not issued until Nov 7) are among the biggest swings (rank 1 of 52 and 5 of 52), but other months reach nearly as far and neither can be attributed.

MONDAY Ask Minnesota's SNAP agency (DCYF) for advance notice of payment gaps and for a de-identified SNAP–shelf linkage pilot. Our data do not show shelves absorbing benefit cuts, so do not plan on it.

BLS CPI-U food at home, Midwest (CUUR0200SAF11), 46 months: Feb 2022–Sep 2025 and Jun–Jul 2026 (BLS published no October 2025 CPI) · Minnesota DCYF SNAP accounting months · The Food Group HouseholdsReg, statewide

Shelves lean toward low-income, low-access tracts, but the residents of those tracts live farther from one.

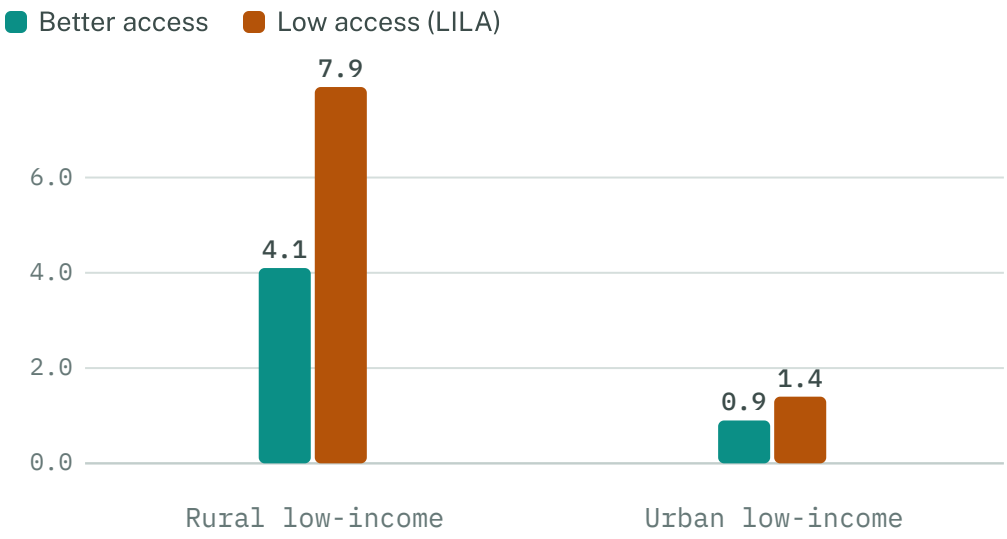
Among low-income tracts, those far from a grocery store are also farther from a shelf: rural residents live 7.9 miles from the nearest one against 4.1 (34% vs 8% beyond 10 miles); urban 1.4 vs 0.9. They are also somewhat less likely to contain a shelf (borderline, p = 0.066).

Across all tracts, 35% of low-income, low-access (LILA) tracts host a reporting shelf, against 24% of other tracts.

Distance helps explain Q1: counties farther from shelves record less activity per resident (ρ −0.34 visits, −0.42 individuals).

Straight-line miles from residents' census blocks to the nearest reporting shelf, population-weighted; not travel time.

Miles from residents to the nearest reporting shelf, low-income tracts



MONDAY Before proposing a building, check the 142 LILA tracts with no shelf (51 counties) for hours, a mobile stop and transit. Extend hours first; start where residents live farthest from a shelf.

USDA ERS Food Access Research Atlas 2019 (LILA: 1 mile urban, 10 miles rural; 2010 tracts crosswalked to 2020) · The Food Group shelves reporting in 2024 · Census 2020 blocks (population, interior points)

Four things The Food Group can do on Monday, each tied to what we found.

ALLOCATE · Q6 · SAUHARDA

Listen first, then plan by service area

Talk first to the 6 queue counties. For supply, add a poverty share to the forecast split at a weight staff choose, by service area.

SITE · Q8 · ROSHIT

Hours and mobile stops before buildings

In LILA tracts with no shelf, extend hours or add a mobile stop first; check with local partners and tribal nations.

ARGUE · Q4/Q5 · BIRUK

Ask for notice and linked data

Advance warning when benefits are late, and a de-identified SNAP–shelf linkage pilot. Our data do not show shelves absorbing cuts.

MEASURE · ALL · JACOB

Fix the reporting, then survey

A month-close flag, a county check for large shelves, and a survey that includes non-users: the test that can reject the score.

ALSO **Daniela:** limits and AI use (appendix A9). August forecast: appendix A1. AI helped write code; every number here is re-derived by a separate script (153 of 153 match).

FoodBridge, our planning tool · USDA TEFAP formula, 7 CFR 251.3(h), and Minnesota DCYF Office of Economic Opportunity TEFAP fact sheet (county poverty and unemployment drive TEFAP shares)

APPENDIX · PULL FORWARD WHEN ASKED

Appendix

A1 Forecast (Q7) · A2 Need-score tests · A3 The 14 flagged counties · A4 Q1 detail · A5 Prices · A6 SNAP and November 2025 · A7 Reach (Q8) · A8 Sources · A9 Limits and data trust

The forecast is a run rate staff can recompute: the last three months per working day, weighted toward July.

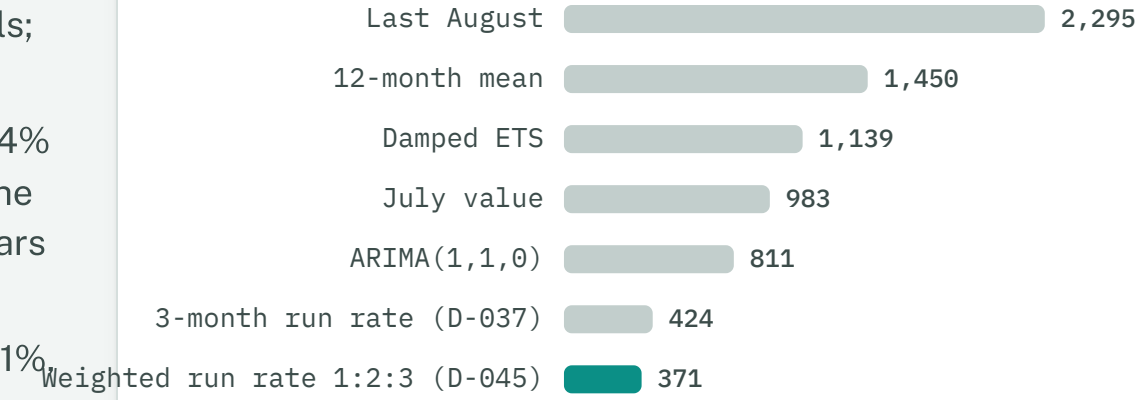
VISITS	POUNDS	CHILDREN	SENIORS	ADULTS
230,620	12,122,0	251,086	82,965	378,592
Households	PoundsReg	ChildrenReg	SeniorsReg	AdultsReg

- Weights 1:2:3 for May, June, July × August's 21 working days (July had 22); ages add to individuals; Wilkin = 0 (no reporting site 2022–26).
- On 37 monthly back-tests it cut county RMSE 3–4% against the equal-weight rate (pounds 0.7%); in the August folds 2–19%, but worse in 8 of 12 field-years of the 2024–25 Augusts.
- Risk: one county carries the error (visits Dakota 41%, pounds Ramsey 69%, individuals Hennepin 54%) and July completeness is unknown. No August 2026 value was used.

Backtest RMSE by candidate (county, sponsor units)

Pooled over the August folds, the organizers' metric. Highlighted: the submitted model.

Visits | Pounds | Children | Seniors | Adults



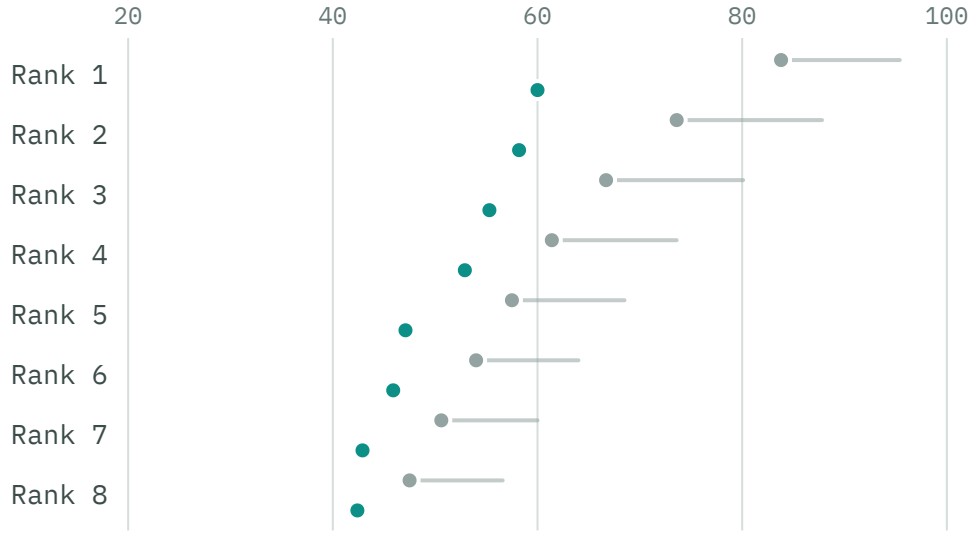
outputs/eval/q7_backtest.csv, q7_origin_check_summary.csv, q7_backtest_by_county.csv · outputs/forecasts/q7_august2026_state.csv · statewide sums over 86 counties · The Food Group org_FSStats 2022-01 to 2026-07

The four addendum items, and how the flagged list compares with chance.

ADDENDUM ITEM	OUR ANSWER
1 · Falsifiable definition	Need = household inability to reliably obtain adequate food, asked for or not. Score = average percentile of food-insecurity rate, poverty rate, no-vehicle share.
2 · Need, not demand	No activity enters the score; it tracks verified SNAP enrolment (ρ 0.66) and tracks shelf activity per resident only weakly (ρ 0.09 to 0.14). The SNAP helpline is too sparse to add a check: the median county logged 2 food-related contacts in 2024 (ρ 0.17, [−0.03, 0.36]).
3 · Evidence to reject	Score does not rise with SNAP participation (held: 0.66), or not holding poverty fixed (narrow: 0.28, [0.02, 0.50]); a survey of non-users shows no hardship difference (not run).
4 · Evidence to revise	Ranks move under weights or inputs (6 of 14 flagged survive all); audits find only reporting gaps (then add a reporting term).

Flagged-county gaps vs a random pairing of need and activity

Largest gaps (percentile points): observed vs a random pairing of need and activity (median to 95th percentile). Observed sits below chance at every rank.



docs/reference/q6_addendum.md · outputs/tables/q6_falsification_checks.csv, q6_permutation_null.csv, q6_activity_correlations.csv · 20,000 permutations

The 14 flagged counties, and which tests each one survives.

FLAGGED COUNTY (ᴹ METRO)	NEED PCTL	ACTIVITY (5)	WEIGHTS (3)	INPUTS (3)	BLOCK COUNTY	CENSUS NOISE
Chippewa	99	✓	3/3	3/3	✓	100%
Lyon	86	✓	3/3	3/3	✓	100%
Wadena	89	✓	3/3	3/3	✓	100%
Todd	78	✓	3/3	3/3	✓	99%
Swift	70	✓	3/3	3/3	✓	98%
Nobles	93	✓	3/3	3/3	✓	94%
Clearwater	90	✗	3/3	3/3	✓	95%
Aitkin	78	✓	2/3	1/3	✓	90%
Benton ᴹ	56	✓	2/3	2/3	✗	82%
Winona	82	✗	1/3	1/3	✓	76%
Carlton ᴹ	60	✗	3/3	3/3	✓	74%
Roseau	55	✓	2/3	2/3	✓	64%
Norman	51	✓	1/3	1/3	✓	52%
Beltrami	98	✗	2/3	1/3	✓	40%

Activity (5): flagged under all five activity universes and cleaning variants. **Weights (3):** flagged under three other weightings of the three inputs. **Inputs (3):** flagged when each input is dropped in turn. **Block county:** flagged when shelves are counted in the county of their census block. **Census noise:** share of 2,000 redraws of poverty and vehicle shares within their margins of error that still flag the county.

Shaded rows pass every test (census noise 90% or more). The rest enter or leave with the weights, the vehicle input or one shelf's county; call them *watch*, not *audit first*.

In those 6, reported pounds reach a median **38%** of the estimated meal gap (64% for all counties): this restates the flag in meals, an upper bound.

12 of 14 flagged counties are outside metro areas.

outputs/tables/q6_robust_set.csv, q6_divergence_sensitivity.csv, geo_sensitivity_q6_divergence.csv · Feeding America MMG 2026, Census ACS 2019–23 (inputs and margins of error)

Lowest and highest activity per resident, and what little the county characteristics explain.

LOWEST, VISITS 2024	RESIDUAL
Scott	−2.94
Pope	−2.28
Nicollet	−2.12
Norman	−2.08
Lac qui Parle	−1.97

HIGHEST	RESIDUAL
Steele	2.92
Washington	2.32
Rice	1.81
Olmsted	1.74
Renville	1.72

Need does not predict either list: these are simply the lowest and highest activity per resident. Scott's rank depends on one shelf's county booking (slide 2); Steele, Washington, Rice and Olmsted do not change.

County characteristics explain little (R^2 0.10–0.18). Households without a vehicle is the steadiest term: positive in 9 of 9 models (standardized 0.12–0.22), but it is also a need input, and the nine models reuse the same counties.

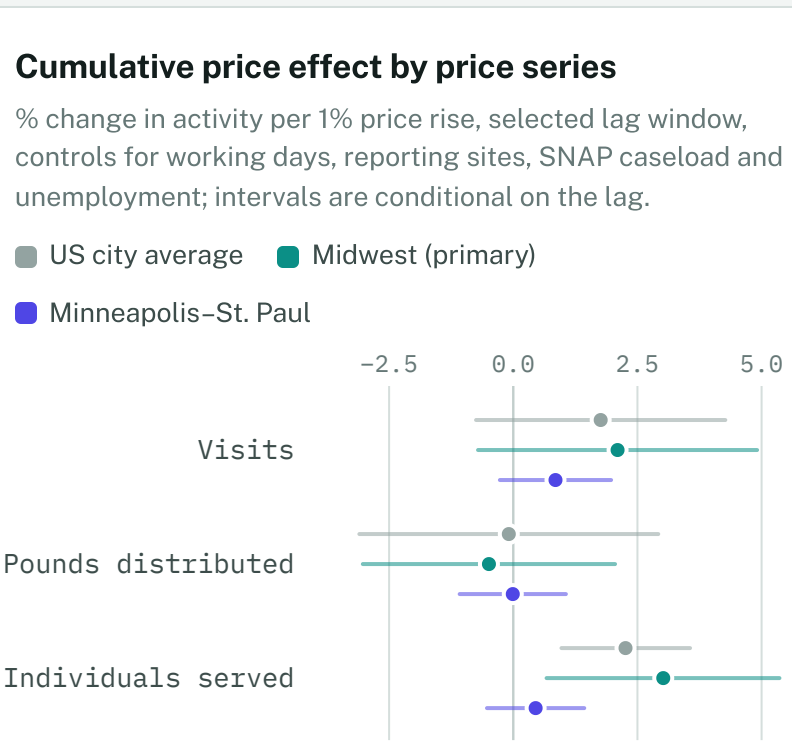
"Regional hub" and "thin network" are hypotheses: no client origins are observed, and Norman, Aitkin and Wadena have more shelves per resident than most counties.

outputs/tables/q1_outliers.csv, q1_explanatory_models.csv, q1_county_concentration.csv, geo_sensitivity_q1_outliers.csv

Three signals, three price series: one weak positive estimate, and it is fragile.

Cumulative price effect by price series

% change in activity per 1% price rise, selected lag window, controls for working days, reporting sites, SNAP caseload and unemployment; intervals are conditional on the lag.



Two of nine intervals exclude zero, both for individuals served. Without February–June 2022 the Midwest estimate is −0.9 [−3.2, 1.5]; from 2023 on it is −5.1 [−10.2, 0.0].

The calendar is the stronger cycle: monthly totals move with working days (elasticity about 0.6). Models use 46 months because BLS published no October 2025 CPI.

Literature: higher food prices are linked to more hardship among SNAP households (Gregory and Coleman-Jensen, USDA ERS 2013/2014), not to a pantry elasticity.

BLS CPI-U food at home CUUR0000SAF11 (US), CUUR0200SAF11 (Midwest), CUURS24ASAF11 (Minneapolis–St. Paul) · outputs/tables/q4_price_models.csv, q4_price_fragility.csv, eda_calendar_effect.csv

SNAP and food shelves move together within county agencies; the two benefit episodes cannot be attributed.

EPISODE VS ORDINARY MONTHS	VISITS	POUNDS	INDIVIDUALS
November 2025 residual (points)	11.3	6.9	11.5
… standard deviations from ordinary	1.5	1.2	1.3
… rank among 52 months	5 of 52	7 of 52	6 of 52
April 2023 residual: standard deviations	2.6	1.6	1.8
… rank among 52 months	1 of 52	4 of 52	4 of 52

What was happening in November 2025: Minnesota DCYF told recipients on Oct 21 that November benefits would not be issued; full issuance came Nov 7 (cards loaded from Nov 8). The same month the federal shutdown was in its second month, SNAP work-rule and energy-deduction changes took effect, and the state announced $4M for food shelves. December fell below its usual level.

The panel link (0.27 per 1%, [0.05, 0.48], 83 units, 54 months) is one signal of three and describes differences between counties' changes, not what a household does when it loses benefits.

November 2025 is within ordinary swings (about 1.5 typical swings); April 2023, the one month beyond 2, is when emergency allotments ended; other changes may coincide. No larger response where SNAP use is higher in either episode; we do not claim a causal effect.

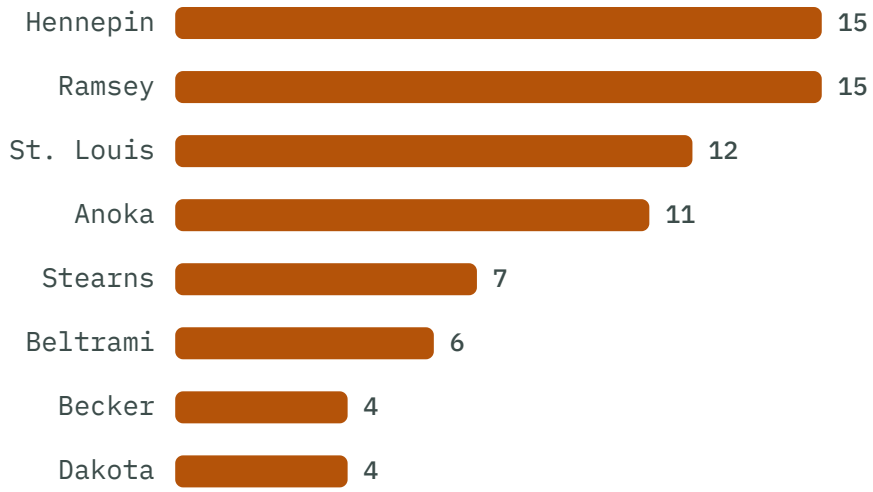
Minnesota DCYF Nov 9 SNAP partner update (catalog 51) and Attorney General release Nov 8 (52) · outputs/tables/q5_noise_summary.csv, q5_panel_coefficients.csv, q5_event_study.csv, q5_nov2025_surprise.csv · docs/reference/policy_timeline.md

The within-income gap, tested three ways, and the counties with the most LILA tracts and no shelf.

LOW-INCOME TRACTS: LOW-ACCESS VS BETTER-ACCESS	ODDS RATIO	
Rural, Fisher exact	—	0.33
Urban, Fisher exact	—	0.16
Both, Mantel–Haenszel	0.71 (0.50–1.02)	0.066
+ tract population	0.67 (0.48–0.94)	0.021

DISTANCE TO THE NEAREST REPORTING SHELF	MILES	BEY
Rural, better access	4.1	8% over 10
Rural, low access	7.9	34% over 10
Urban, better access	0.9	29% over 1
Urban, low access	1.4	52% over 1

Counties with the most LILA tracts lacking a shelf



USDA ERS Food Access Research Atlas 2019, 2010 tracts crosswalked to 2020 (10 tracts under 90% overlap) · outputs/tables/q8_size_adjusted.csv, q8_county_footprint_gaps.csv · county-clustered logistic regression, 562 low-income tracts

A8 APPENDIX · DATA SOURCES, ALL CITED

Four sources came with the challenge; these additional sources do the explaining.

ADDITIONAL SOURCE (PUBLISHER, SERIES)	GRAIN	VINTAGE	USED FOR
U.S. Census Bureau · ACS 5-year: B17001 poverty, B19013 income, B01003 population, B08201 vehicles	County (87); tract only through USDA flags	2019–2023 five-year	Q1 Q6 Q8
USDA FNS · SNAP Data Tables, state-by-month cross-check	State	Monthly, to 2026-06	Q5
USDA ERS · Food Access Research Atlas: LILA tract flags	Census tract, 2010 geography	2019 data, 2021 release	Q8
USDA ERS · Rural-Urban Continuum Codes 2023	County	2023 codes on 2020 population	Q1 Q5 Q6
U.S. BLS · Local Area Unemployment Statistics, county unemployment rate	County	Monthly, 2022-01 to 2026-07	Q4 Q5 Q7
U.S. Census Bureau · 2020–2010 tract relationship file, Minnesota	Tract pair with shared land area	2020 vs 2010 geography	Q8
USDA ERS · SNAP Retailer Access Map (SRAM), retailer sensitivity check	Census tract, 2020 geography	2025 retailers, 2020–2024 ACS	Q8
U.S. Census Bureau · 2020 Redistricting Data (P.L. 94-171): block population and interior points	Census block (198,705 in Minnesota)	2020 census	Q1 Q8
FEMA, CDC SVI, EIA gasoline, HUD FMR, Census PEP · context pack, checked for a usable signal (catalog 26–30)	County, state	2021–2026	Q1 Q6
Minnesota DCYF, DHS, MDE, USDA and others · dated policy timeline (22 sourced events)	State, federal	2021–2026	Q4 Q5

Provided: Feeding America Map the Meal Gap, BLS food-price data, Minnesota SNAP enrolment, Minnesota school-meal data. Full catalog with publisher, series, grain, vintage, access date and reason chosen: sources.html and docs/sources/catalog.md (67 rows). Map the Meal Gap citations: Dewey et al. 2024; Ribar et al. 2025 (revised 28 Jul 2026), 2026.

What these results are not, how the data were cleaned, and what would change our answers.

Not causal	Associations and event contrasts. No causal policy effect is identified.
Not unmet people	Visits and individuals repeat across months; no unique-person count, no household adequacy; need is inferred, not observed.
Not in the numbers	Tribal nations' own food programs and tribal-agency SNAP, shelves that report late or not at all, and barriers such as language, work schedules and hours.
Q6 screening	A queue for audit; the survey test is designed, not run; the poverty overlap with Feeding America's own model is disclosed.
Would change it	Linked household data, client origins and hours, a representative county survey, the sponsor's county booking for large shelves.

ROWS

28,261

site × month, invalid months dropped

WINDOW

2022-01 → 2026-07

no August 2026 value used

Statewide food-shelf network, classified on the organization master, history keeps inactive sites, six diagnosed keying cells repaired by mechanism, a blank month is never a zero, July 2026 completeness cannot be measured. August 2026 has one fewer working day than July.

Each result carries a reproducible check: 153 of 153 numbers on these slides match an independent re-derivation; 58 decisions are logged.